

Causal Inference Project Summary

This quarter, I participated in a study of causal inference methods with my mentor Qian Meng. Over the course of 10 weeks, we were able to cover chapter 1 and parts of chapters 2 and 3 of the textbook *A First Course in Causal Inference* by Peng Ding. We covered topics such as the Yule-Simpsons paradox, potential outcomes, completely randomized experiments, the Fisher Randomization Test (FRT), the Neyman repeated sampling inference, and some discussion about observational studies. The majority of our time was spent writing proofs and occasionally using R to replicate analyses from the text.

The textbook starts with a chapter detailing the Yule-Simpson paradox, a statistical paradox where trends within groups contradict the overall trend. One of the most famous examples includes the 1973 UC Berkeley admissions data where men had a significantly higher overall admission rate, despite women having higher department-based admission rates in several departments. We then moved onto potential outcomes and the completely randomized experiment (CRE), which addresses the Yule-Simpsons paradox by ensuring that treatment assignment is independent of potential outcomes, allowing unbiased estimation of causal effects without confounding from lurking variables. Potential outcomes outline a set of definitions and assumptions necessary for modeling for causal relationships. These assumptions allow us to effectively measure causal relationships between treatment groups and outcomes.

We then discussed how the exact FRT p-value and p-value by Monte Carlo are calculated using the assumption that the only source of randomness is in the assignment variable. I conducted my own p-value calculation on the LaLonde dataset in RStudio. At the end of the project, I did more proofs regarding Neyman's inference and the difference in means test statistic. We touched on some applications of causal inference techniques on observational data but unfortunately ran out of time to fully delve into the topic. The basic idea for applying causal inference is to take into consideration possible confounding variables that affect the treatment assignment randomness assumption.

I had a great time learning about causal inference, which was a completely new topic to me when I began this project. I appreciate the passion my mentor expressed throughout the quarter, Qian was an amazing mentor with a real passion for statistics.